XIANRUI HOU

Tianjin, China | +86 186 9805 9388 | houxianrui@mail.dlut.edu.cn
hmartians587.github.io

EDUCATION

Dalian University of Technology (Project 985), China Sep. 2023 – Jun. 2027 (expected)
Electronic Information Engineering (English Intensive) Average: 83.6/100
School of Information and Communication Engineering

Selected coursework (out of 100): Artificial Intelligence and Information Society (96), Human Intelligence (**100**), Probability and Statistics A (95), Computer Principles (Bilingual) (95), Discrete Mathematics (92), Electronic Measurement Technology (96), Communication Principles Laboratory (96), Signal Processing Laboratory II: System Analysis and Design (**99**), Electromagnetic Fields and Microwave Technology Laboratory (**100**).

RESEARCH INTERESTS

Multimodal learning; software engineering; continual learning; computer vision; representation learning; LLMs and multimodal agents.

RESEARCH EXPERIENCE

IIAU Lab, Dalian University of Technology Feb. 2026 – Sep. 2026
Undergraduate Researcher; Group lead: Prof. Huchuan Lu
Supervisor: Shuai Yu (Postdoctoral Researcher; Assistant Professor, Donghua University)

HIM-CoT: Reasoning over the Human-Instrument-Music Chain for Musical Performance Deepfake Detection

Co-first author *Under review at ICLR 2027*

- Co-designed HIM-CoT to assess Human–Instrument–Music consistency using structured music, motion, and performer–instrument contact evidence for deepfake detection.
- Constructed HIM-DF with audio and visual forgeries; selected generation models and built CoT supervision datasets to ground reasoning in audio-visual evidence.
- Co-developed the experimental pipeline and ran selected detection experiments; evaluated structured evidence and cross-modal discrepancies in instrumental performances.
- Research outcomes: HIM-CoT reached 89.3% accuracy, 95.0% AP, and 94.2% AUC on HIM-DF; evaluation on the LAV-DF benchmark yielded 90.6% accuracy.

ENCODE Lab, Westlake University Aug. 2026 – Present
Undergraduate Researcher; Supervisor: Huan Wang, Assistant Professor

Online Distribution Shift Awareness in Continual Learning

Sole first author *Research in progress; target: CVPR 2027*

- Investigate distribution-change awareness during continual training without task boundaries, focusing on reliable detection as the model's representations evolve.
- Lead method development and implementation for distinguishing new distributions from recurring states; build the online training and evaluation pipeline under faculty supervision.
- Design evaluation streams with abrupt shifts, gradual transitions, and recurrence; use multi-seed runs, module ablations, and event-level analysis to assess reliability and detection delay.

Learning What to Execute and When: Reasoning Protocols for LLM Inference

Co-author *Under review at ICLR 2027*

- Reproduced selected LLM inference-time reasoning baselines and compiled comparative results; assessed answer accuracy against mean model calls per question.
- Research outcomes: REPROT learns protocols that select and compose reasoning operations; the paper reports 84.2% accuracy at 3.9 model calls per question across eight benchmarks.

PUBLICATIONS AND MANUSCRIPTS

HIM-CoT: Reasoning over the Human-Instrument-Music Chain for Musical Performance Deepfake Detection

Shuai Yu*, **Xianrui Hou***, Yue Zhu, Yuecheng Zhuo, Li Wang, Lijun Wang, Yongqi Shan, Huchuan Lu.

*Equal contribution. *Under review at ICLR 2027*

Learning What to Execute and When: Reasoning Protocols for LLM Inference

Yu Huo, Kun Zeng, Siyu Zhang, Haoyue Liu, Yuecheng Zhuo, **Xianrui Hou**, YuQuan Lu, Xiaoying Tang.

Under review at ICLR 2027

PROFESSIONAL EXPERIENCE

Panasonic Software Development (Dalian) Co., Ltd. Jun. 2026 – Jul. 2026

Intern

Raspberry Pi-Based In-Vehicle Safety and AI Companion System

- Trained an emotion-recognition model and designed the GUI for a Raspberry Pi driver-monitoring system covering identity, fatigue, and facial-expression states.
- Built a Raspberry Pi driver-monitoring pipeline with MediaPipe eye and mouth landmarks; combined a finite-state machine with smoothed dynamic thresholds for fatigue detection.
- Integrated YuNet and SFace for 1:N driver identification with face extraction and alignment; exported a trained mouth-risk binary classifier to ONNX for Raspberry Pi inference.
- System integration: FER outputs use five-sample voting; fatigue and negative-emotion events trigger DeepSeek responses and asynchronous edge-tts/pygame speech playback.

SELECTED PROJECTS

Plantar-Pressure-Based Motion Direction Prediction

Core Member, Undergraduate Innovation Project Jun. 2025 – Mar. 2026

- Designed the sensor-acquisition circuit using EDA tools; preprocessed plantar-pressure signals in Python with imputation, filtering, outlier handling, and sliding-window segmentation.
- Developed an LSTM with temporal attention to predict motion direction from plantar-pressure sequences, using recurrent encoding to capture gait dynamics.

HONORS AND AWARDS

National Finals Second Prize May 2026

National Undergraduate Communication System Design Competition; leader of a two-member team.

Provincial Third Prize May 2025

Lanqiao Cup National Software and Information Technology Competition for University Students.

LEADERSHIP

Class Organizing Committee Member, Dalian University of Technology

Class distinctions: Exemplary Class for Academic Atmosphere (2025; 10 classes selected university-wide annually), and Excellent Class for Academic Atmosphere (2024).

TECHNICAL SKILLS

Programming: Python, C, MATLAB, VHDL.
Deep learning: PyTorch; custom training loops, multi-GPU training, checkpointing.
Development: Linux/Bash, Git, LaTeX; remote experiment management, modular design.
Edge computing and vision: Raspberry Pi, ONNX, MediaPipe, YuNet, SFace.
Hardware and simulation: FPGA, ModelSim SE, Simulink, Multisim, Proteus, JLC EDA.

LANGUAGES

Mandarin Chinese; English: CET-6 (482). IELTS examination planned for early November 2026.